

KHABO-KOI

Restaurant Discovery, Reservation and Management Platform

Software Engineering Project Report

Course: Software Engineering Lab (CSE 3422)

Prepared by

Abu Nasir Patwary (011213015)

Mohammad Shafin Ahmed (011213058)

Md. Rakibul Alam (011182070)

Mst. Sahanaj Parvin (011213105)

Md. Shahariar Hossen Ridoy (011201466)

Submitted to

Samin Sharaf Somik

Lecturer, Department of CSE, UIU

A complete web application for finding restaurants, booking tables, pre-ordering food and managing operations through role-specific portals.

Contents

1. Introduction
2. Project Description
3. Technology Used
4. Future Works
5. Conclusion

1. Introduction

1.1 Short Description of the Project

Khabo-Koi is a complete restaurant discovery, reservation and operations-management platform. Customers can search restaurants, view branches on a map, reserve a table, track bookings, pre-order food and get dining tips based on the data. Restaurant Managers, Branch Managers and Platform Admins use separate protected portals to manage day-to-day operations, access role-scoped analytics and control users, restaurants, assignments, reservations, tables, menus and pre-orders.

The application has a React front-end, Django REST API, Django ORM and PostgreSQL back-end architecture. The system is built on actual working relationships: one restaurant can have several branches, each branch has tables, customers make bookings for certain date and time slots, and food pre-orders remain associated with those reservations.

1.2 Objectives

- Create a single source for finding restaurants, cuisines, branches and food items.
- Allow authenticated users to reserve tables and maintain a private booking history.
- Support food pre-ordering linked to a reservation, including amount and payment-status tracking.
- Allow customers to use an artificial intelligence-based dining assistant that can tailor restaurant and food suggestions to their preferences, location and budget.
- Show customers an interactive map to find restaurant branches and explore available food outlets by area.
- Assign Restaurant Managers to a set of restaurants and limit their access to unrelated businesses.
- Provide Branch Managers with a specific work area for a designated branch, including reservations, tables, menu availability, pre-orders and notifications.
- Provide Platform Admins access to users, restaurants, manager assignments, bookings and operational history.
- Deliver role-specific analytics for platform, restaurant and branch decision-making.
- Use secure authentication, permission checks, automated tests and Git-based cooperation to enhance reliability.

1.3 Motivation

Restaurant discovery and booking are commonly spread across telephone calls, social media pages and separate payment or ordering systems. Customers might not know which branch has suitable tables, while restaurant staff do not have a common view of bookings and pre-orders. Khabo-Koi was developed to meet the requirements of customer convenience and functional operational control.

The project also exemplifies fundamental software-engineering techniques: layered architecture, relational data modelling, RESTful APIs, role-based access control, parallel Git development, automated testing and incremental integration.

2. Project Description

2.1 List of Features

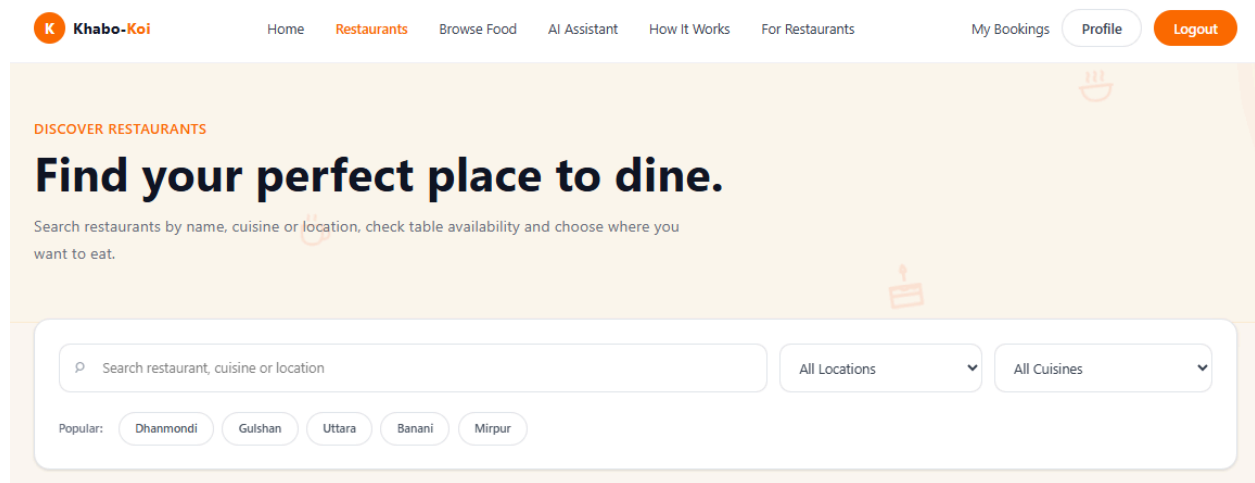
- Restaurant Discovery
- Interactive Map
- Table Reservation
- Food Pre-order
- My Bookings
- AI Dining Assistant
- Authentication and Role-Based Access Control
- Platform Admin Portal

- Restaurant Manager Portal
- Branch Manager Portal
- Analytics
- Notifications

2.2 Description of Features

2.2.1 Restaurant Discovery

Khabo-Koi features a searchable and filterable interface to help customers find restaurants. Users can search by restaurant name, cuisine or location and sort the results according to the available options. Restaurant cards display useful information such as ratings, cuisines, branches and table capacity, helping customers compare restaurants before opening the detailed reservation page.



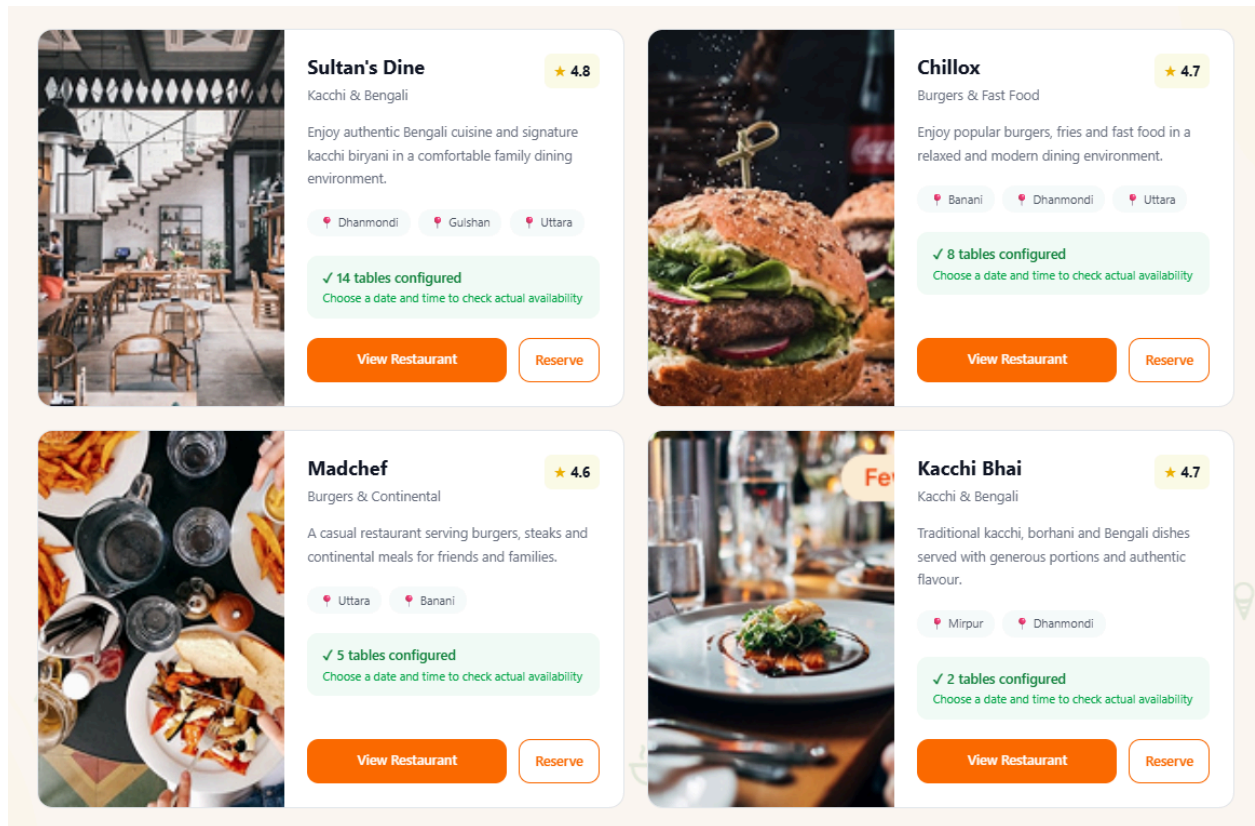


Figure 1. Restaurant Discovery.

2.2.2 Interactive Map

The platform also has an interactive map that shows restaurant branches in Dhaka. It uses OpenStreetMap data, Leaflet and React-Leaflet. Customers can use the map to locate restaurant branches and explore restaurants by location.

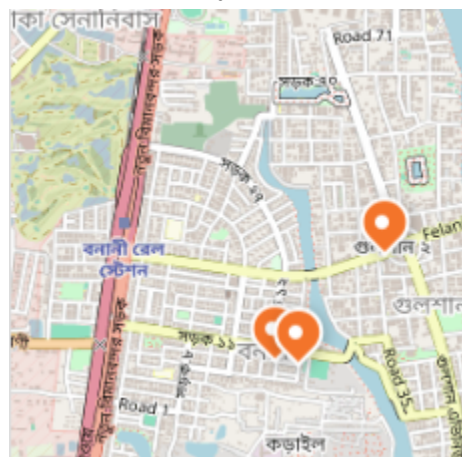




Figure 2. Searchable restaurant discovery interface with integration of a map of Dhaka.

2.2.3 Table Reservation

An authenticated customer can reserve a table at a chosen restaurant branch. A reservation records the branch, table, reservation date, start time, end time and number of guests. The system checks whether a table is available before creating a booking and keeps booking statuses such as PENDING, CONFIRMED, COMPLETED and CANCELLED.

SMART TABLE BOOKING

Reserve your table

Select your branch, date, time, party size and seating preference.

Branch

Dhanmondi



Reservation Date

mm/dd/yyyy



Arrival Time

Select Time



Number of Guests

2 Guests



Seating Preference

Any Seating

Indoor

Outdoor

Window Side

Customer Information

Login-based customer information will replace this later.

Your name

Phone number

Check Table Availability

REVIEW RESERVATION

Table available

Booking Summary

Restaurant	Sultan's Dine	Branch	Dhanmondi
Date	2026-10-08	Time	6:00 PM
Guests	2 Guests	Seating	Any Seating
Selected table	Table T1	Capacity	Up to 2 guests

Table reservation

Advance payment

Total now

0

Required only when food is added to the reservation.

0

You can pre-order food in the next step, or reserve only the selected table.

Special Request

Add an optional note for the restaurant.

+ Add Special Request

Figure 3. The customer can make a reservation according to their requirements.

2.2.4 Food Pre-order

Once a table reservation is made, customers can place orders for food and drinks that can be prepared for their arrival. The selected items, quantities, unit prices, total amount, advance payment amount, payment status and preparation or operational status are stored in the system. This links food orders directly with the customer's table booking.

YOUR ORDER

Pre-order Summary

Mutton Kacchi Biryani

₹580

1 × ₹580

Table reservation

₹0

Food subtotal

₹580

Order total

₹580

Advance Payment

50% required for food pre-order

Food preorder total

₹580

Pay now (50%)

₹290

Remaining at restaurant

₹290

Pay ₹290 Advance

Pay 50% now to confirm your food pre-order. The remaining amount will be paid at the restaurant.

Confirm Food Pre-order

Skip Food Pre-order

Your table reservation is already confirmed.

SECURE PAYMENT

Pay ₹290 Advance

Complete the 50% advance payment required for your food pre-order.

Payment Method

bKash

Nagad

Card

Mobile Number

01700000000

Food total	₹580
Paying now	₹290
Remaining	₹290

Your payment details are used to confirm the advance amount for this preorder.

Confirm Secure Payment ₹290

YOUR ORDER

Pre-order Summary

Mutton Kacchi Biryani ₹580
1 × ₹580

Table reservation ₹0
Food subtotal ₹580

Order total ₹580

Advance Payment

PAID

50% required for food pre-order

Food preorder total ₹580
Pay now (50%) ₹290
Remaining at restaurant ₹290

✓ Advance payment successful

Method: BKASH
Transaction: KK-PAY-30-150254

Pay 50% now to confirm your food pre-order. The remaining amount will be paid at the restaurant.

Confirm Food Pre-order

Skip Food Pre-order

Your table reservation is already confirmed.

✓ Reservation Confirmed

RESTAURANT

Sultan's Dine

Dhanmondi

Your table is booked.

Booking #30 has been saved to your Khabo-Koi account.

DATE

October 8, 2026

TIME

6:00 PM

GUESTS

2

TABLE

T1

FOOD PRE-ORDER

Pre-order Summary

✓ 50% Advance Paid

Mutton Kacchi Biryani

1 × ₳580

₳580

Food total

₳580

Advance paid

₳290

Remaining at restaurant

₳290

PAYMENT METHOD

BKASH

TRANSACTION ID

KK-PAY-30-150254

View My Bookings

Explore Restaurants

Figure 4. Customers can pre-order food to reduce preparation time.

2.2.5 My Bookings

The My Bookings feature enables customers to see their personal reservation history. Customers can view details about the restaurant, branch, booking date, booking time, number of guests, assigned table and booking status. This provides an easy-to-use option for users to keep track of their reservations from one place.

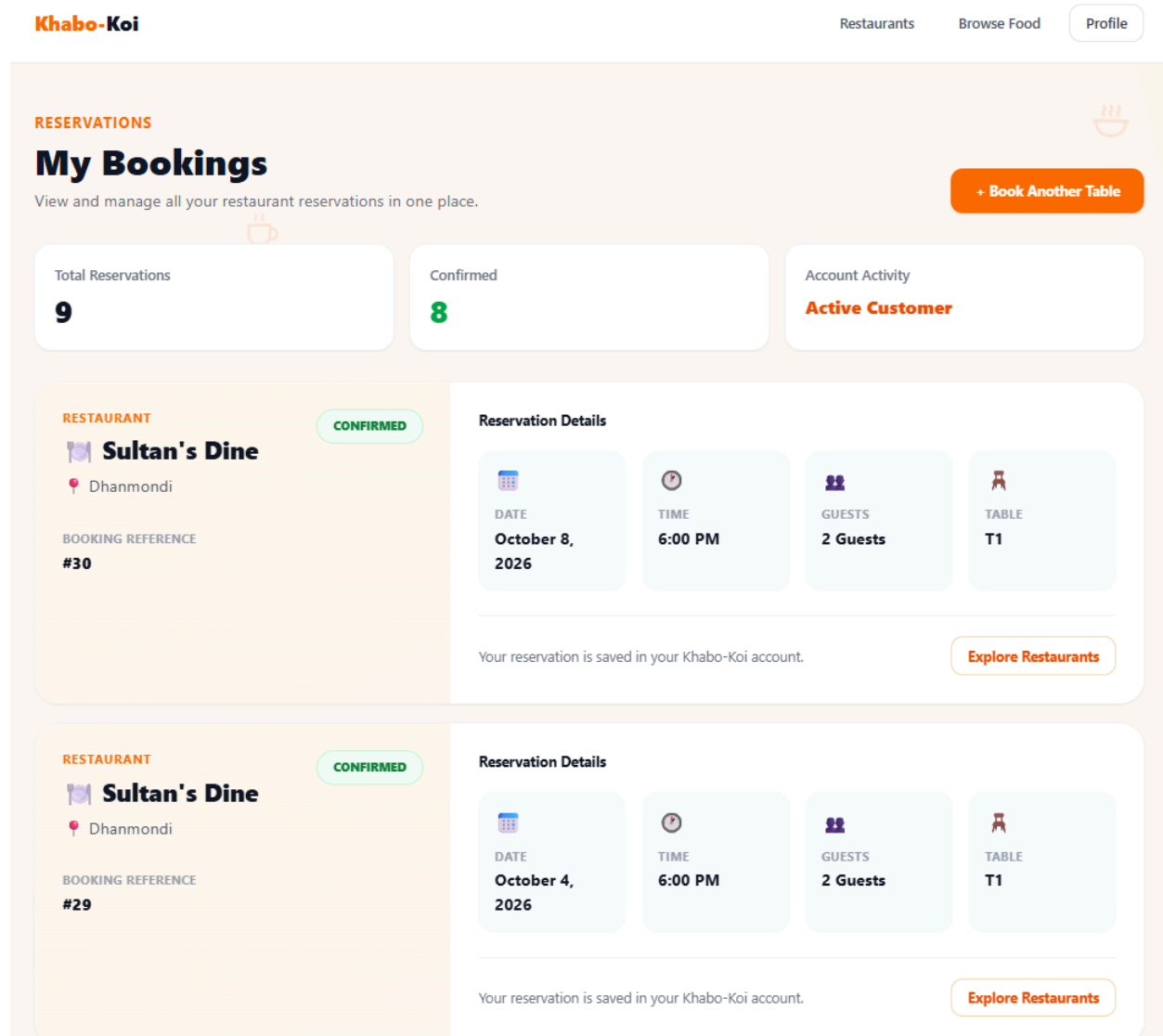


Figure 5. Customers can view all their booking information.

2.2.6 AI Dining Assistant

Khabo-Koi offers an AI Dining Assistant that can handle natural-language queries about restaurants, food choices, locations and budgets. The assistant provides suggestions using the restaurant, branch and menu information available in the Khabo-Koi system. If the external AI service is unavailable, the system can fall back to a local database-based service so that recommendations continue to be based on real project data.

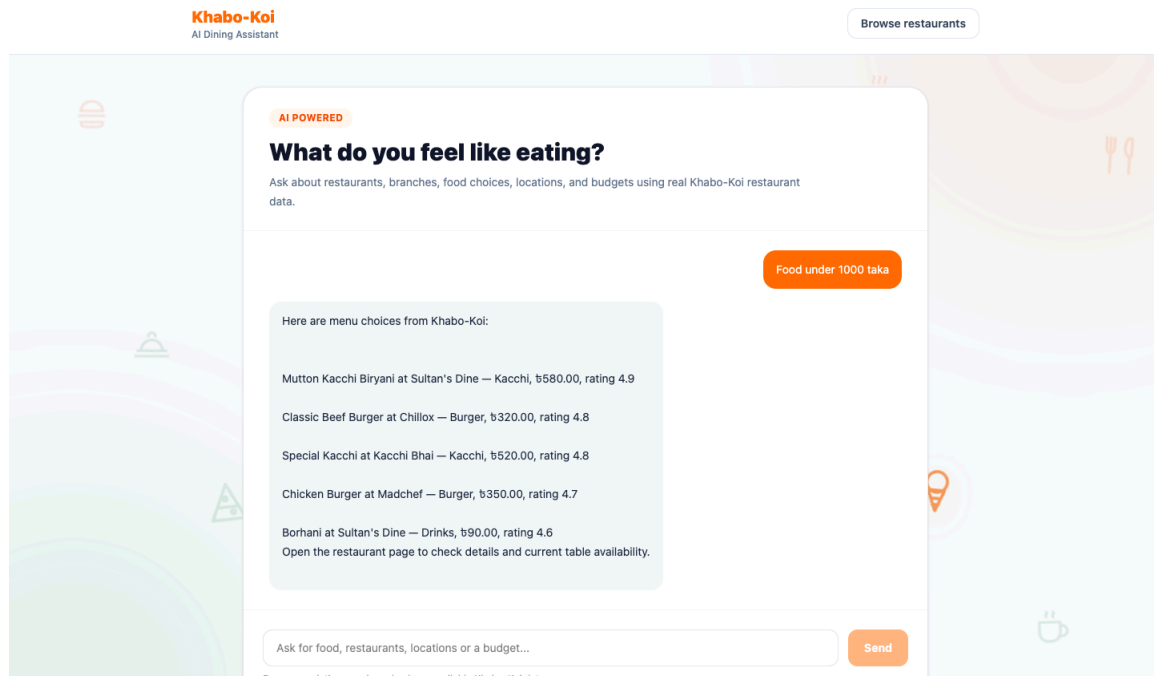


Figure 6. Data-driven dining suggestions based on Khabo-Koi records.

2.2.7 Authentication and Role-Based Access Control

The platform implements secure authentication and role-based authorization to regulate access to individual parts of the system. Django handles user authentication and password hashing, while JSON Web Tokens are used for authenticated API sessions. Users are assigned roles such as Customer, Restaurant Manager, Branch Manager or Platform Admin, and backend permission checks ensure that only data and operations authorized for a particular role are available to that role.

2.2.8 Platform Admin Portal

The Platform Admin Portal provides a platform-wide management and monitoring. Platform Admins have access to users, account status, restaurants, Restaurant Manager assignments, Branch Manager assignments, bookings, analytics and operational history management. An admin check is done on the server to prevent unauthorized access to admin operations.

Khabo-Koi

PLATFORM ADMINISTRATION

Secure operations portal

Dashboard

Restaurants

Users

Restaurant Managers

Branch Managers

Bookings

Operational History

System Settings

100%

PLATFORM INTELLIGENCE

Admin Dashboard

Signed in as selenium-admin

Platform at a glance

Monitor platform growth, compare restaurants and branches, and review persisted booking and preorder activity.

Reporting period

Last 30 days

Apply period

Total restaurants

4

All registered restaurants

Active restaurants

4

Currently active on the platform

Total users

10

All customer and staff accounts

Active users

10

Accounts currently permitted to sign in

PERFORMANCE REPORT

Bookings and preorder analytics

2026-09-09 to 2026-10-08

Export CSV

Reservations

20

42 reserved guests

Completion rate

15.0%

3 completed

Cancellation rate

5.0%

1 cancelled

Preorder value

BDT 7,220

10 food preorders

Amount collected

BDT 3,610

Recorded payments and advances, not verified gateway revenue

Outstanding

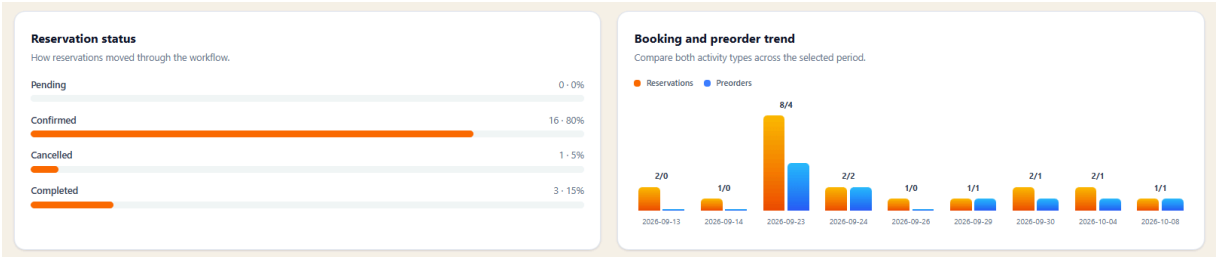
BDT 3,610

Remaining preorder amount

Branch performance

Compare bookings, guests and recorded preorder value.

RESTAURANT	BRANCH	BOOKINGS	COMPLETED	GUESTS	PREORDER VALUE
Sultan's Dine	Dhanmondi	16	3	34	BDT 6,380
Kacchi Bhai	Mirpur	1	0	2	BDT 520
Chillox	Banani	3	0	6	BDT 320
Madchef	Banani	0	0	0	BDT 0
Chillox	Dhanmondi	0	0	0	BDT 0
Kacchi Bhai	Dhanmondi	0	0	0	BDT 0
Sultan's Dine	Gulshan	0	0	0	BDT 0
Chillox	Uttara	0	0	0	BDT 0
Madchef	Uttara	0	0	0	BDT 0
Sultan's Dine	Uttara	0	0	0	BDT 0



Top-selling food items

Ranked by ordered quantity in persisted preorders.

1	Mutton Kacchi Biryani	Kacchi	11 sold	BDT 6,380
2	Special Kacchi	Kacchi	1 sold	BDT 520
3	Classic Beef Burger	Burger	1 sold	BDT 320

K

Khabo-Koi

PLATFORM ADMINISTRATION

Secure operations portal

D

Dashboard

R

Restaurants

U

Users

RM

Restaurant Managers

BM

Branch Managers

B

Bookings

H

Operational History

S

System Settings

soon

selenium-admin

selenium-admin@example.com

Sign out

PLATFORM ADMINISTRATION

Restaurant Oversight

Restaurant operations

Review every registered restaurant and control whether it is available across the Khabo-Koi platform.

Refresh restaurants

Live

4

Registered restaurants

Live

4

Active restaurants

Live

10

Restaurant branches

Live

1

Active Manager access

Search restaurant or cuisine

All restaurants

C

Chillox

Active

Burgers & Fast Food - Rating 4.7

3 branches - 1 food items - 0 active Managers

Deactivate

K

Kacchi Bhai

Active

Kacchi & Bengali - Rating 4.7

2 branches - 1 food items - 0 active Managers

Deactivate

M

Madchef

Active

Burgers & Continental - Rating 4.6

2 branches - 1 food items - 0 active Managers

Deactivate

S

Sultan's Dine

Active

Kacchi & Bengali - Rating 4.8

3 branches - 1 food items - 1 active Managers

Deactivate

Deactivating a restaurant hides it from customers and disables its Manager assignments. Reactivation does not restore old Manager access.

K

Khabo-Koi

PLATFORM ADMINISTRATION

Secure operations portal

D

Dashboard

R

Restaurants

U

Users

RM

Restaurant Managers

BM

Branch Managers

B

Bookings

H

Operational History

S

System Settings

soon

selenium-admin

selenium-admin@example.com

Sign out

PLATFORM ADMINISTRATION

User Management

Users and access

Search accounts, manage product roles, and suspend or reactivate access. Every change is validated again by Django.

Refresh users

Live

10

Total users

Live

1

Restaurant Managers

Live

1

Platform Admins

Live

0

Suspended accounts

Search by username or email

All roles

All statuses

USER	STATUS	PRODUCT ROLE	JOINED	ACTIONS
A AbuNasirPatwary nasirchester145@gmail.com	Active	Customer Save	9/1/2026	Suspend
B branchmanager No email provided	Active	Branch Manager Save	9/23/2026	Suspend
D ditom2026 ditom2026@tst.com	Active	Customer Save	9/1/2026	Suspend
H Habib habib@gmail.com	Active	Customer Save	9/13/2026	Suspend
H Himel himel@gmail.com	Active	Customer Save	9/23/2026	Suspend
H HimeTest3 himehaz@gmail.com	Active	Customer Save	9/23/2026	Suspend

K

Khabo-Koi

PLATFORM ADMINISTRATION

Secure operations portal

D

Dashboard

R

Restaurants

U

Users

RM

Restaurant Managers

BM

Branch Managers

B

Bookings

H

Operational History

S

System Settings

100%

selenium-admin

selenium-admin@example.com

Sign out

PLATFORM ADMINISTRATION

Manager Access

Restaurant assignments

Give active Restaurant Managers access to the restaurants they operate. Deactivated records remain visible as access history.

Refresh access

Live

1

Assignment records

Live

1

Active assignments

Live

1

Assigned Managers

Live

1

Managed restaurants

NEW ACCESS

Assign a Manager

Restaurant Manager

Choose a Manager

Restaurant

Choose a restaurant

Assign access

Search Manager, email, or restaurant

All access

MANAGER	RESTAURANT	ASSIGNED BY	STATUS	ACTION
selenium-manager selenium-manager@example.com	Sultan's Dine	selenium-admin	Active	Deactivate

K

Khabo-Koi

PLATFORM ADMINISTRATION

Secure operations portal

D

Dashboard

R

Restaurants

U

Users

RM

Restaurant Managers

BM

Branch Managers

B

Bookings

H

Operational History

S

System Settings

100%

selenium-admin

selenium-admin@example.com

Sign out

PLATFORM ADMINISTRATION

Branch Manager Access

Branch Manager

Choose user

Branch

Choose branch

Assign branch

Search manager, restaurant, or branch

All statuses

MANAGER	RESTAURANT / BRANCH	STATUS	ACTION
branchmanager	Sultan's Dine Diamond	Active	Deactivate

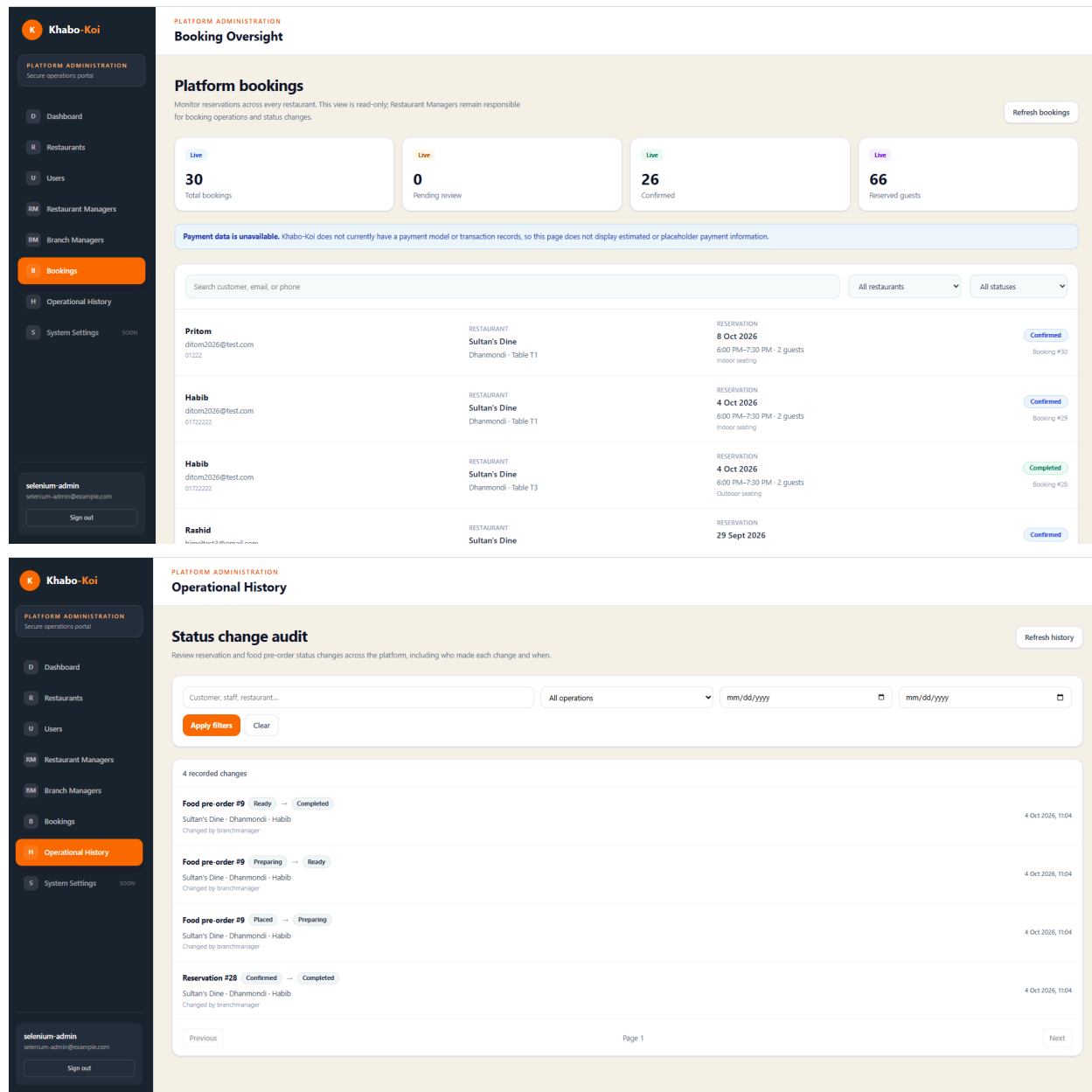


Figure 7. Platform-Admin's workflow

2.2.9 Restaurant Manager Portal

A protected workspace is provided for Restaurant Managers which only allows them to access their current restaurant assignments. They are able to add information about restaurants, keep their branches, tables and menus up to date, and filter reservation queues and move bookings through approved status changes. Authorized branches can be compared using analytics which includes comparing reservations, number of guests, completion/cancellation rates, popular items and recorded pre-order value.

K

Khabo-Koi

RESTAURANT MANAGEMENT
Assigned operations portal

Dashboard

Restaurant Profile

Reservations

Tables

Menu

Branches

Operational History

selenium-manager

selenium-manager@example.com

Sign out

RESTAURANT PERFORMANCE

Manager Dashboard

Welcome, selenium-manager

Compare branches and monitor bookings, guests, menu demand and recorded preorder value across your assigned restaurants.

Reporting period

Last 30 days

Apply period

Assigned restaurants

1

Active Platform Admin assignments

Active branches

3

3 total branches

Active tables

14

14 total tables

Available menu items

1

1 total items

PERFORMANCE REPORT

Bookings and preorder analytics

2025-09-09 to 2025-10-08

Export CSV

Reservations

16

34 reserved guests

Completion rate

18.8%

3 completed

Cancellation rate

6.3%

1 cancelled

Preorder value

BDT 6,380

8 food preorders

Amount collected

BDT 3,190

Recorded payments and advances, not verified gateway revenue

Outstanding

BDT 3,190

Remaining preorder amount

Reservation status

How reservations moved through the workflow.

Booking and preorder trend

Compare both activity types across the selected period.

K

Khabo-Koi

RESTAURANT MANAGEMENT
Assigned operations portal

Dashboard

Restaurant Profile

Reservations

Tables

Menu

Branches

Operational History

selenium-manager

selenium-manager@example.com

Sign out

RESTAURANT MANAGEMENT

Restaurant Profile

Public restaurant information

Update only the restaurant details visible to customers. Rating and platform activation remain protected fields.

Refresh profile

S

Active

Rating 4.8

Sultan's Dine

Kacchi & Bengali

Restaurant name

Sultan's Dine

Cuisine

Kacchi & Bengali

Description

Enjoy authentic Bengali cuisine and signature kacchi biryani in a comfortable family dining environment.

Image URL

https://example.com/restaurant.jpg

Discard changes

Save restaurant profile

K

Khabo-Koi

RESTAURANT MANAGEMENT
Assigned operations portal

D

Dashboard

R

Restaurant Profile

B

Reservations

T

Tables

M

Menu

L

Branches

H

Operational History

selenium-manager

selenium-manager@example.com

Sign out

RESTAURANT MANAGEMENT

Reservations

Reservation queue

Review bookings for assigned restaurants and move them through the approved status workflow.

Customer or phone

All statuses

All dates

mm/dd/yyyy

Apply filters

CUSTOMER	LOCATION	SCHEDULE	GUESTS	STATUS	ACTIONS
Pritom 01222	Sultan's Dine Dhamond - Table T1	2026-10-08 18:00:00-19:30:00	2	CONFIRMED	COMPLETED CANCELLED
Habib 0172222	Sultan's Dine Dhamond - Table T3	2026-10-04 18:00:00-19:30:00	2	COMPLETED	Final
Habib 0172222	Sultan's Dine Dhamond - Table T1	2026-10-04 18:00:00-19:30:00	2	CONFIRMED	COMPLETED CANCELLED
habib 01705167229	Sultan's Dine Dhamond - Table T2	2026-09-30 18:00:00-19:30:00	2	CANCELLED	Final
Rifat 01700	Sultan's Dine Dhamond - Table T3	2026-09-30 18:00:00-19:30:00	2	COMPLETED	Final
Rashid 01255	Sultan's Dine Dhamond - Table T1	2026-09-29 18:00:00-20:30:00	2	CONFIRMED	COMPLETED CANCELLED
Pretty 0135	Sultan's Dine Dhamond - Table T1	2026-09-26 18:00:00-20:30:00	2	COMPLETED	Final
Himel 1234	Sultan's Dine Dhamond - Table T2	2026-09-24 18:00:00-20:30:00	2	CONFIRMED	COMPLETED CANCELLED
Himel 01722	Sultan's Dine Dhamond - Table T1	2026-09-24 18:00:00-19:30:00	2	CONFIRMED	COMPLETED CANCELLED
himel	Sultan's Dine Dhamond - Table T2	2026-09-23 21:00:00-23:00:00	2	CONFIRMED	COMPLETED CANCELLED

K

Khabo-Koi

RESTAURANT MANAGEMENT
Assigned operations portal

D

Dashboard

R

Restaurant Profile

B

Reservations

T

Tables

M

Menu

L

Branches

H

Operational History

selenium-manager

selenium-manager@example.com

Sign out

RESTAURANT MANAGEMENT

Tables

Seating inventory

Manage bookable tables inside branches you own.

Add table

Branch

Sultan's Dine - Dhamond

Table number

Capacity

2

Seating

Indoor

☒ Active

Create table

Search table, branch, or seating type

TABLE	BRANCH	CAPACITY	SEATING	STATUS	ACTIONS
T1	Sultan's Dine Dhamond	2	INDOOR	Active	Edit Deactivate
T2	Sultan's Dine Dhamond	4	INDOOR	Active	Edit Deactivate
T3	Sultan's Dine Dhamond	4	OUTDOOR	Active	Edit Deactivate
T4	Sultan's Dine Dhamond	6	WINDOW	Active	Edit Deactivate
T5	Sultan's Dine Dhamond	2	INDOOR	Active	Edit Deactivate
T6	Sultan's Dine Dhamond	2	INDOOR	Active	Edit Deactivate
...	Sultan's Dine	...	...	...	Edit Deactivate

K

Khabo-Koi

RESTAURANT MANAGEMENT

Assigned operations portal

D

Dashboard

R

Restaurant Profile

B

Reservations

T

Tables

M

Menu

L

Branches

H

Operational History

selenium-manager

selenium-manager@sampleit.com

Sign out

Menu catalogue

Maintain real menu items for the restaurants assigned to you.

Add menu item

Name

Category

Price

Image URL

Description

Available

Create menu item

Search menu item, category, or restaurant

M

SULTAN'S DINE - KACCHI

Mutton Kacchi Biryani

No description

Available

Edit

Make unavailable

:\$80.00

K

Khabo-Koi

RESTAURANT MANAGEMENT

Assigned operations portal

D

Dashboard

R

Restaurant Profile

B

Reservations

T

Tables

M

Menu

L

Branches

H

Operational History

selenium-manager

selenium-manager@sampleit.com

Sign out

Branches

Branch locations

Create, update, and safely deactivate locations belonging to your assigned restaurants.

Add a branch

Name

Phone

Address

Latitude

Longitude

Opening time

Closing time

Active

Create branch

Search branch, restaurant, or address

SULTAN'S DINE

Dhanmondi

Dhanmondi, Dhaka

11:00-23:00

Edit

Deactivate

Active

SULTAN'S DINE

Gulshan

Gulshan, Dhaka

11:00-23:00

Edit

Deactivate

Active

SULTAN'S DINE

Uttara

Active

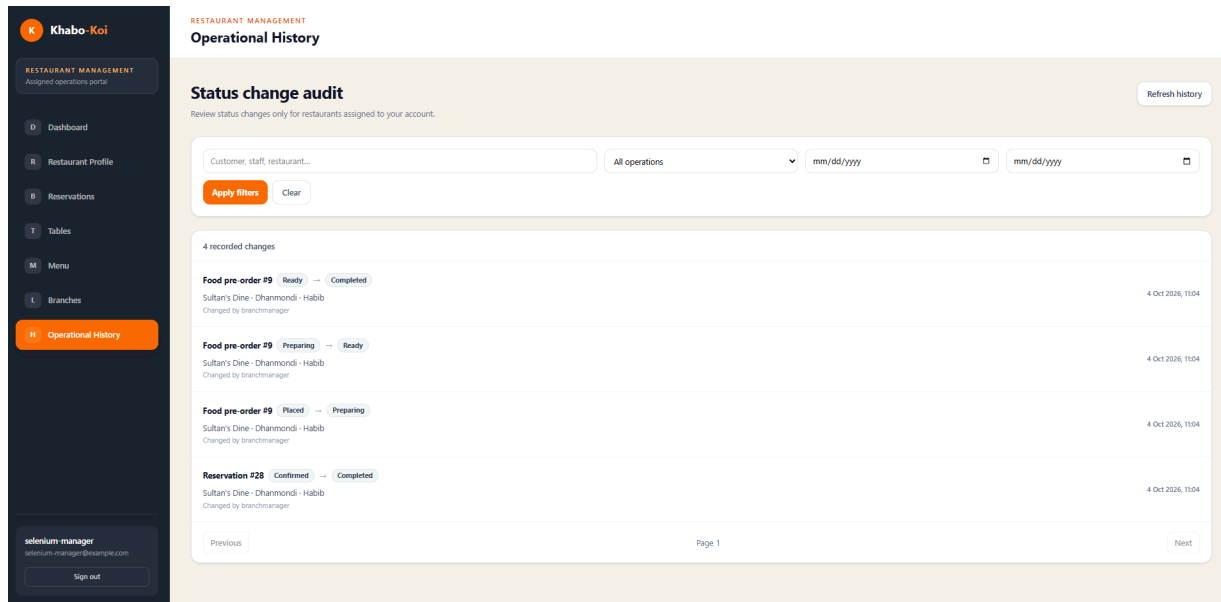
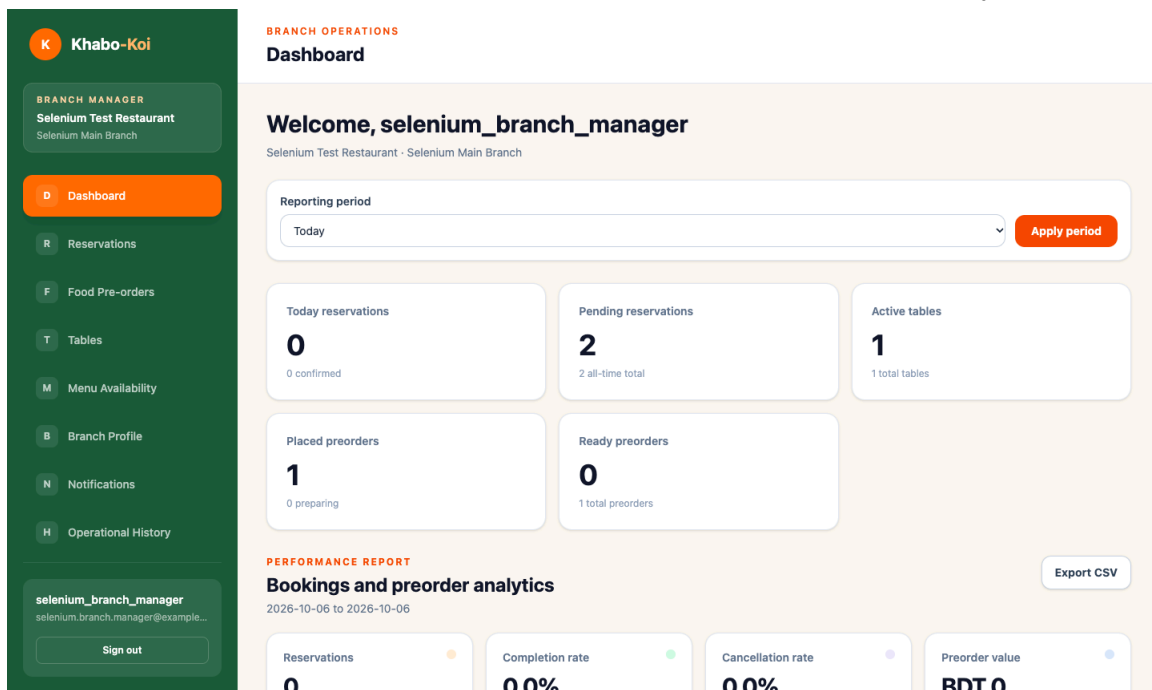


Figure 8. Restaurant Manager's workflow

2.2.10 Branch Manager Portal

The Branch Manager Portal is a portal that is reserved for the day to day management of a specific branch to which they are assigned. The Branch Manager is able to manage reservations, food pre-orders, tables, menu, branch information, notifications and operational history. They also have a dashboard where they will find details for each of their branches: daily reservations, activities that await, tables that are occupied, pre-orders and analytics.



K

Khabo-Koi

BRANCH MANAGER

Sultan's Dine

Dhamondi

D

Dashboard

R

Reservations

F

Food Pre-orders

T

Tables

M

Menu Availability

B

Branch Profile

N

Notifications

H

Operational History

BRANCH OPERATIONS

Dashboard

Welcome, branchmanager

Sultan's Dine - Dhamondi

Reporting period

Today

Apply period

Today reservations

1

15 confirmed

Pending reservations

0

19 all-time total

Active tables

6

6 total tables

Placed preorders

3

2 preparing

Ready preorders

0

8 total preorders

PERFORMANCE REPORT

Bookings and preorder analytics

2026-10-08 to 2026-10-08

Export CSV

Reservations

1

2 reserved guests

Completion rate

0.0%

0 completed

Cancellation rate

0.0%

0 cancelled

Preorder value

BDT 580

1 food preorders

Amount collected

BDT 290

Recorded payments and advances, not verified gateway revenue

Outstanding

BDT 290

Remaining preorder amount

Reservation status

How reservations moved through the workflow.

Pending

0 - 0%

Confirmed

1 - 100%

Cancelled

0 - 0%

Completed

0 - 0%

Booking and preorder trend

Compare both activity types across the selected period.

Reservations

Preorders

1/1

2026-10-08

Top-selling food items

Ranked by ordered quantity in persisted preorders.

1

Mutton Kacchi Biryani

Kacchi

1 sold

BDT 580

K

Khabo-Koi

BRANCH MANAGER

Sultan's Dine

Dhamondi

D

Dashboard

R

Reservations

F

Food Pre-orders

T

Tables

M

Menu Availability

B

Branch Profile

N

Notifications

H

Operational History

branchmanager

Authorized branch access

Sign out

BRANCH OPERATIONS

Reservations

Customer or phone

All statuses

All dates

Apply filters

CUSTOMER	SCHEDULE	TABLE	GUESTS	STATUS	ACTIONS
Pritom 01222	2026-10-08 18:00:00-19:30:00	T1	2	CONFIRMED	COMPLETED CANCELLED
Habib 01722222	2026-10-04 18:00:00-19:30:00	T1	2	CONFIRMED	COMPLETED CANCELLED
Habib 01722222	2026-10-04 18:00:00-19:30:00	T3	2	COMPLETED	
habib 01705167229	2026-09-30 18:00:00-19:30:00	T2	2	CANCELLED	
Rifat 01700	2026-09-30 18:00:00-19:30:00	T3	2	COMPLETED	
Rashid 01255	2026-09-20 19:00:00-20:30:00	T1	2	CONFIRMED	COMPLETED CANCELLED
Pretty 01255	2026-09-26 19:00:00-20:30:00	T1	2	COMPLETED	
Himel 1234	2026-09-24 19:00:00-20:30:00	T2	2	CONFIRMED	COMPLETED CANCELLED
Himel 017222	2026-09-24 18:00:00-19:30:00	T1	2	CONFIRMED	COMPLETED CANCELLED
himel 01255	2026-09-23 21:30:00-23:00:00	T2	2	CONFIRMED	COMPLETED CANCELLED
Farhin 01722	2026-09-23 21:30:00-23:00:00	T1	2	CONFIRMED	COMPLETED CANCELLED

K

Khabo-Koi

BRANCH MANAGER

Sultan's Dine

Diamond

D

Dashboard

R

Reservations

F

Food Pre-orders

T

Tables

M

Menu Availability

B

Branch Profile

N

Notifications

H

Operational History

branchmanager

Authorized branch access

Sign out

BRANCH OPERATIONS

Food Pre-orders

ALL

PLACED

PREPARING

READY

COMPLETED

CANCELLED

BOOKING #20

Pritom

2026-10-08 - 18:00:00 - Table T1

Mutton Kacchi Biryani × 1

Order total
\$580.00

Advance
\$290.00

PLACED

PREPARING

CANCELLED

BOOKING #21

Habib

2026-10-04 - 18:00:00 - Table T1

Mutton Kacchi Biryani × 1

Order total
\$580.00

Advance
\$290.00

COMPLETED

Special request
add some extra mayo

BOOKING #22

Rashid

2026-09-29 - 19:00:00 - Table T1

Mutton Kacchi Biryani × 1

Order total
\$580.00

Advance
\$290.00

COMPLETED

Special request
extra mayooo

BOOKING #23

Piku

2026-09-23 - 18:00:00 - Table T2

Mutton Kacchi Biryani × 1

Order total
\$580.00

Advance
\$290.00

PLACED

Special request
Faculty demo special request

PREPARING

CANCELLED

BOOKING #24

Rifat

PREPARING

BOOKING #19

himel

COMPLETED

K

Khabo-Koi

BRANCH MANAGER

Sultan's Dine

Diamond

D

Dashboard

R

Reservations

F

Food Pre-orders

T

Tables

M

Menu Availability

B

Branch Profile

N

Notifications

H

Operational History

branchmanager

Authorized branch access

Sign out

BRANCH OPERATIONS

Tables

Add table

Table number

2

Indoor

Add table

Search table, seating, or capacity

INDOOR

Table T1

Capacity: 2 guests

Active

Edit

Deactivate

INDOOR

Table T2

Capacity: 4 guests

Active

Edit

Deactivate

OUTDOOR

Table T3

Capacity: 4 guests

Active

Edit

Deactivate

WINDOW

Table T4

Capacity: 6 guests

Active

Edit

Deactivate

INDOOR

Table T5

Capacity: 2 guests

Active

Edit

Deactivate

INDOOR

Table T6

Capacity: 2 guests

Active

Edit

Deactivate

K

Khabo-Koi

BRANCH MANAGER

Sufan's Dine

Branch Manager

D

Dashboard

R

Reservations

F

Food Pre-orders

T

Tables

M

Menu Availability

B

Branch Profile

N

Notifications

H

Operational History

branchmanager

Authorized branch access

Sign out

BRANCH OPERATIONS

Menu Availability

Branch-local availability

The Restaurant Manager owns the master menu. You control only whether each existing item is available at this branch.

KACCHI

Mutton Kacchi Biryani

₹580

Available at branch

Search menu

K

Khabo-Koi

BRANCH MANAGER

Sultan's Dine

Dhanmondi

D

Dashboard

R

Reservations

F

Food Pre-orders

T

Tables

M

Menu Availability

B

Branch Profile

N

Notifications

H

Operational History

branchmanager

Authorized branch access

Sign out

BRANCH OPERATIONS

Branch Profile

SULTAN'S DINE

Dhanmondi

Update operational contact details and hours. Branch identity stays under Restaurant Manager control.

Phone

Address

Opening time

Closing time

Save branch details

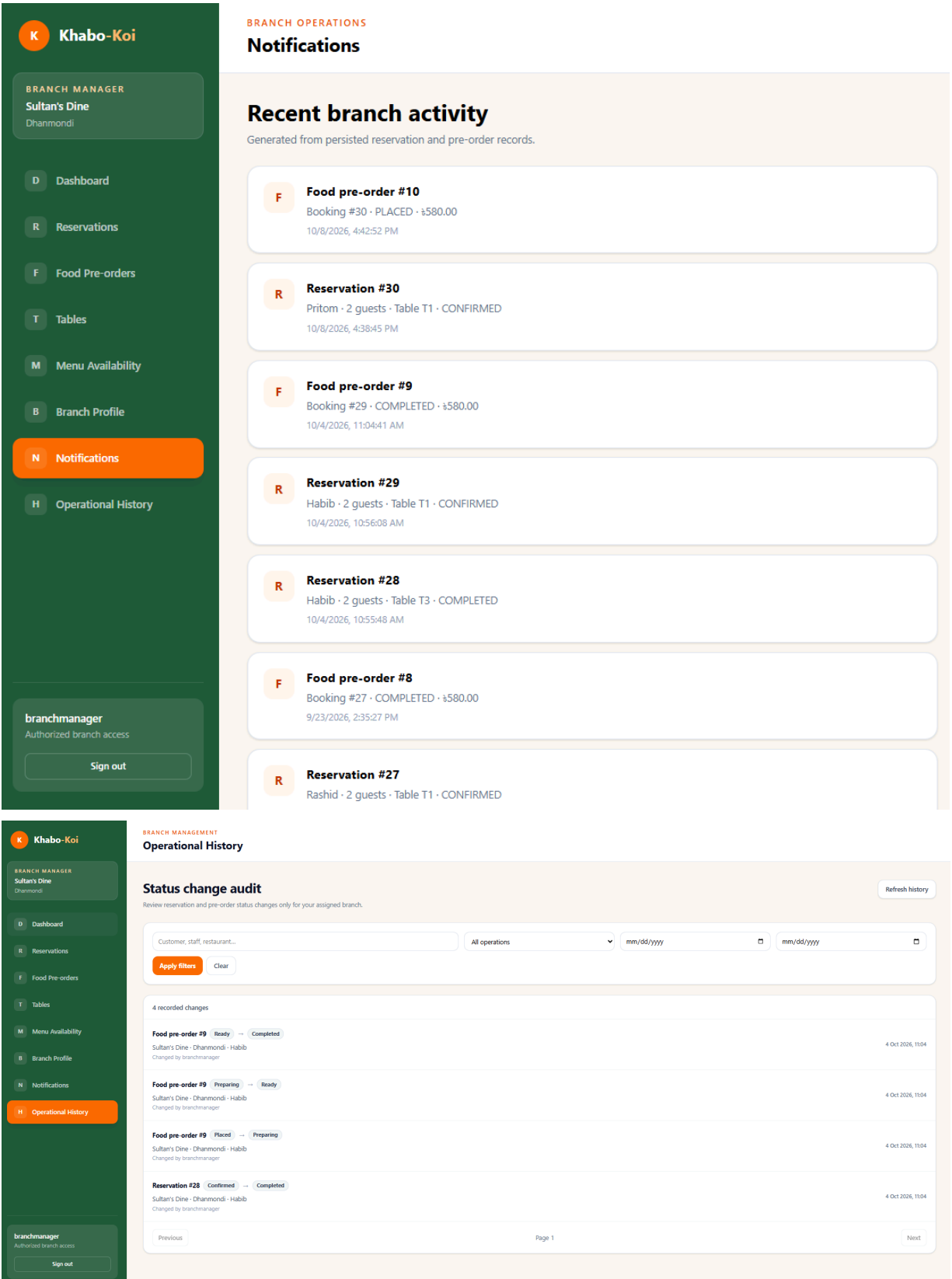


Figure 9. Branch Manager's workflow

2.2.11 Analytics

Khabo-Koi offers analytics per role for Platform Admins, Restaurant Managers and Branch Managers. The analytics provide information such as reservation totals, guests, booking status breakdowns, completion and cancellation percentage, pre-order values, collected values, outstanding values, popular food items, time based trends and branch performance. It also has features for more in-depth analysis such as date filters and CSV export.

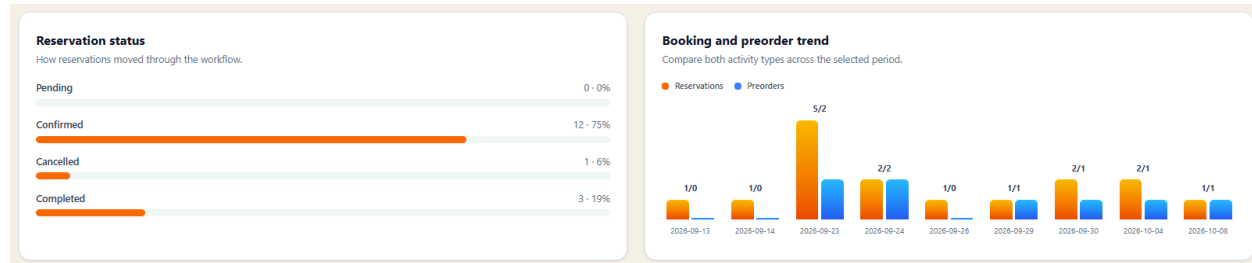


Figure 10. Analytical data

2.2.12 Notifications

The notification feature enables Branch Managers to be aware of all the important activities related to the branch and operational changes. Operational Status History and notifications (provided in the Branch Manager workspace) make it easier to keep track of and review important events and changes.

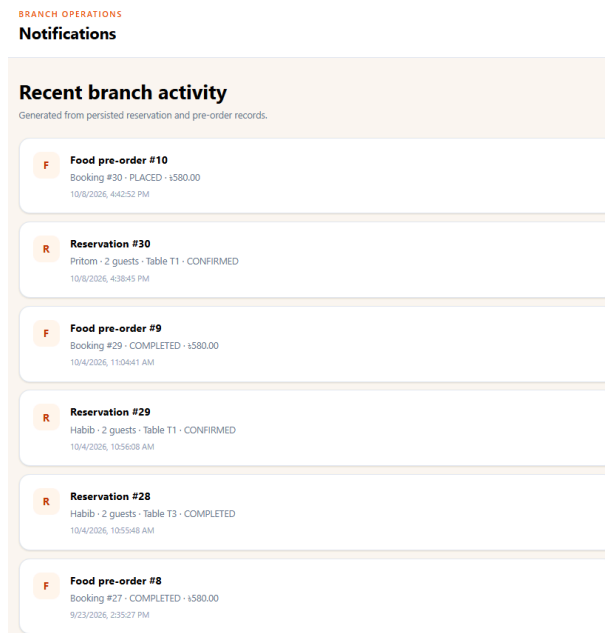


Figure 11. The warning and confirmation of action to take.

3. Technology Used

3.1 Frontend

React 19: constructs UI using reusable components, and state-driven pages.

Vite 8: includes the development server and module bundling and optimized production builds.

Tailwind CSS 4: adds responsive layouts, typography, cards, forms and role-specific themes to the framework.

React Router 7: Introduces Protected route wrappers and customer/routes/Admin/routes/Restaurant Manager/routes/Branch Manager.

Leaflet and React-Leaflet: Create interactive map of Dhaka to draw restaurant boughs.

Structuring AI-assistant responses with formats in the conversation interface safely: React Markdown.

3.2 Backend

Implementations in the business layer, ORM Models and Migrations, validation and server-side authorization are done in python and django 6.1. The Django REST Framework is used to serve the JSON APIs that are used by the react frontend. Separate views and permission classes enforce the customer, restaurant, branch and platform scopes.

3.3 Database

Persistent users, profiles, restaurants, branches, menu items, tables, bookings, pre-orders, assignments, notifications and operational history are stored in PostgreSQL. Foreign keys are used to establish relations, unique constraints to keep no invalid duplicate and Django migrations to handle schema changes between team branches.

3.4 AI Model and Dining Assistant

The AI Dining Assistant is based on the Groq API provided by OpenAI through an openai compatible client and model openai/gpt-oss-20b. The system passes the restaurant, branch and menu context data from Khabo-Koi to enable recommendations that are based on project data. If the external AI service is not available, it can provide an alternative to a database search with the added benefit of giving useful recommendations.

3.5 Authentication and Authorization

The Django's User model is handling identity and generating hashes of the password. The API's are authenticated using JSON Web Tokens and refreshed using refresh token. Product roles are placed in UserProfile, Operational scope in RestaurantManagerAssignment and BranchManagerAssignment. By default public registration is set to Customer and elevated roles are only assignable via protected endpoints on the Admin.

Other controls are password validation, login/registration throttling, password reset, email-verification state, protected frontend routes and server-side ownership checks. However, at the front end, the limitations increase usability, and at the back end it's the permissions that are the true security wall.

3.6 REST APIs

The frontend communicates with the Django REST API endpoints for login/authorization, restaurant profiles, restaurant discovery, availability, restaurant booking, food pre-order, AI suggestions, analytics and management operations. The Serializers check the data of requests and responses, and the Query Filter can be used to search, specify a range of roles, select the status of the requests, date criteria, restaurant criteria, branch criteria and table criteria.

3.7 Testing and Quality Assurance

Django's testing framework was used to test the back-end for models, serializers, APIs, permissions, statuses, analytics, authentication, security controls and demo data to ensure they behave as expected. All tests were performed on a PostgreSQL test database to ensure that it works correctly.

There are unit tests for analytics queries, status normalization, CSV export and pagination in the frontend. Again, the project was created using Vite and, once again, it's very successful. In a real browser, Selenium tests were performed on the login page of the Customer, Platform Admin, Restaurant Manager and Branch Manager roles, and on some of the most important workflow pages.

3.8 Git and Team Collaboration

The branch based development process was used which was based on Git and GitHub. Members created feature branches related to authentication, administration, restaurant operation, customer functions, analytics, AI, maps, reminders, and testing and user-interface enhancements. Work was reviewed and merged in a controlled fashion maintaining a readable history, and minimising exposure to the main branch from direct risk.

GitHub Repository: <https://github.com/AbuNasirPatwary/Khabo-Koi.git>

3.9 Overall Architecture

The code is structured into layers: the React front-end interacts with the Django REST API which uses the Django ORM to connect to the PostgreSQL database, and the optional external AI services are accessed via the back-end.

- React user interface
- The Django REST API is used to handle the data: The data will be handled with the Django REST API:
- The data is stored in a relational database Django ORM and PostgreSQL

- The option for external AI service with the possibility of a local fallback, based on a database
- This is the automated verification segment of the Django, Node and Selenium.

4. Future Works

4.1 challenges encountered and solutions taken.

- Different data and file models: Different team members did work on different features and sometimes modified related data and file models. This led to the merging and migration of Git conflicts. To mitigate this, the team developed feature branches, small commits and rigorously reviewed migrations and tested between pushes and pulls of feature changes.
- Different access, data security: Access to different parts of the system needed for different users. For example, a Branch Manager should be able to view only his/her assigned Branch and a Restaurant Manager should be able to view only his/her assigned restaurants. This was mitigated via JWT authentication, role based permissions, manager-assignment models and backend access checks.
- Not supported external AI credentials: During a demo, the Groq key or service might not be available. A fall back plan using the database restores some of the functions but does not create new restaurants or menu items.
- Browser tests were impacted by changing interfaces: The button label and re-rendering of React. Stable routes help to mitigate this problem, as do isolated test users, frequent updates to selectors, and explicit waits.
- Analytics needed suitable data: In case there weren't enough bookings or pre-orders in the analytics database, the analytics pages had insufficient data. Sample and test records were created for meaningful analytics and demonstrations.
- Database migration problems: Sometimes models changed in the database, causing discrepancies between the model in the live code and a pre-existing test database. The issues were addressed by re-creating the test database and using the latest Django migrations.
- Frontend performance: The production version has a relatively large JavaScript bundle. In the future, lazy loading, dynamic imports and code splitting could enhance initial loading performance.

4.2 Planned Upgrades and New Features

- Use SSLCommerz or another secure payment service for all real transactions, callbacks, refunds and reconciliation.
- Remind people of reservations by email, SMS or push notification prior to the reservation time.
- Use WebSocket live updates to avoid manual refreshing of booking and kitchen / pre-order updates.

- Enhance analytics with peak hour forecasting, demand prediction, revenue comparisons and downloadable scheduled reports.
- Add customer reviews and ratings moderation, favorite and personal recommendations from consented history.
- Improve AI by leveraging improved retrieval, conversation history, safety monitoring and model-quality assessment, while maintaining database grounding.
- Support Restaurant self-onboarding with admin approval, documentation and flexible booking policies.
- Upgrade production readiness with HTTPS, environment specific configuration, background jobs, Redis caching, database backup, logging and monitoring and container deployment.
- Make all pull requests and branches more accessible, more responsive-device coverage, performance budgets, and continuous integration.

5. Conclusion

Khabo-Koi is a complete restaurant discovery, reservation and management solution. Customer features include restaurant discovery, interactive map browsing, table booking, food pre-ordering, restaurant assistance using AI, and access to protected management portals for Restaurant Managers, Branch Managers and Platform Admins.

It's running on a PostgreSQL database with its own Django REST APIs and role-based access control, generating analytics and automated testing to ensure reliable and organized operations. The one thing that makes the most sense to the user, is that they can easily find and reserve restaurants, and that authorised employees can easily access the tools they need in order to manage their restaurant data, branches, reservations and other actions on the platform.

The developed system is also a solid base to build upon when adding features such as real payment integration, real-time notifications, more advanced analytics and deployment to a production environment.